

ST.MARY'S HIGH SCHOOL-MBALE.

UGANDA ADVANCED CERTIFICATE OF EDUCATION.

DEPARTMENT OF PHYSICS.

P510/1: PHYSICS.

PRE-UNEB-2026.

Time: 2 hours and 40 minutes

Instructions:

*(i); Attempt **four** questions in total. Each question carries **20 marks**.*

*(ii); You must attempt **one** question from each of the four sections:*

(iii); Section A is compulsory.

(iv); Use the provided data and support materials to guide your solutions.

(v); Clearly show all calculations and assumptions.

SECTION A.

Item 1

The display screens of digital devices use Light Emitting Diodes (LEDs), which emit high amounts of blue light. Scientific research has shown that long exposure to such light is harmful to the human eyes. An ICT student of

senior five in a certain school is worried that their eyes may be at a risk due to continuous use of computers.

Support Material Provided:

- *Wavelength of blue light = 450 nm*
- *Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$,*
- *speed of light in vacuum, $c = 3.0 \times 10^8 \text{ m/s}$*
- *Damage threshold for the human eye = 3.1 eV*

Tasks:

(a); Use the information provided to dispel fears of the student.

(b); Explain to the student some of the symptoms of eye defects which can arise from prolonged exposure to radiations of energy beyond certain threshold.

(c); Based on principles of Physics, suggest possible protective strategies which the student can employ to safeguard his/her eyes.

SECTION B

Item 2

In a hilly district in Uganda, a landslide buried part of a school compound during the rainy season. The Ministry of Education and Sports is seeking practical proposals from physics students to help schools in the district to construct retaining structures that can prevent future landslides effects in schools in the district and other districts prone to landslides.

Support Material Provided:



Slope inclination is 45° , mass of clay soil swept by rain is 3000 kg with density $1,800 \text{ kg m}^{-3}$. Rain increases soil mass by 25% and Coefficient of friction, $\mu = 0.45$.

Tasks:

(a); With evidence, explain what happens when the soil gets saturated.

(b); Propose with justification, the materials that can be used to construct a realistic vertical retaining structures in the district that can be used to protect the classrooms.

Item 3

Many rural areas experience frequent motorcycle accidents due to sharp curves and poor road design. Local council authorities seek physics-based input on safe road redesign.

Support Material Provided:



Radius of curved road = 12 m, Banking angle = 5° , Coefficients of friction: 0.6 (dry), 0.35 (wet). Mass of motorcycle + rider = 180 kg.

Tasks:

(a); With justification, advise the local council on the maximum safe speed that the riders should move at, on such roads under wet and dry conditions.

(b); Suppose that the curved road was not banked, provide convincing advice to the local council authorities, the design that would enable the motorcyclists reach their destinations earlier in both scenarios.

SECTION C

Item 4

Many rural households rely on firewood for cooking, leading to deforestation and health problems. In a bid to save our environment, the government recommended that alternative energy sources like solar be used. In order to demonstrate to the community members, a solar cooking technology with Solar irradiance of 800 W/m^2 , having cooking pot area of 0.2 m^2 and efficiency of 35% is used to boil 2 kg of water from 25°C to 100°C . The community members wondered how this technology works and how long it would take to boil the given mass of water.



Tasks:

- (a); Help the community members understand how the technology works and show to them how long it would take the water to boil.
- (b); Suggest design improvements, based on thermal principles, that can cause the water to boil in a shorter time than in (a) above.

Item 5

A new rural health center is built in a valley where mobile phone signals are weak. You are part of a student team challenged to design a light-based communication system between the health center and a village watch tower located 150 meters away on a hill, in order to ease communication at the health centre. Hint: refractive indices of glass and air are 1.5 and 1.0, respectively.

Tasks:

- a) Explain how communication occurs from the watch tower to the health center through the medium identified.
- b) Suggest **practical improvements** that the team can implement to increase clarity or strength of the signal at the health center.

SECTION D

Item 6

Market vendors have over the years complained of destructive fire

outbreaks caused by overloaded electric power extension cables. These fire outbreaks have often left market vendors devastated due to loss of property. To alleviate this situation, the chairman of the market vendors' association requested you, an A'level student of physics, to evaluate and recommend safer electrical practices.

During your evaluation, you find out that: the market is wired using an electrical cable of length 20m of resistance $0.5 \Omega \text{ m}^{-1}$ carrying a current of 8A for a duration of 4 hours per day. Maximum safe energy dissipation by the cable is 331.2 kJ m^{-1} .

Tasks:

- (a) Analyze the information provided and suggest the cause of fire in the market.
- (b) Propose practical safety improvements and awareness strategies to the vendors to avoid such occurrence.

Item 7

A farming community bordering a game reserve regularly suffers crop destruction due to wild animals entering their fields at night. To address this, they intend to develop a solar-powered electronic alarm system that uses a magnetic sensor to detect the motion of animals. They heard that such a system operates on alternating current (A.C) converted from solar energy. When activated, the system triggers a light-emitting diode (LED) to display the animal detection waveform on a Cathode Ray Oscilloscope (CRO), which has a Y-sensitivity of 20 Vcm^{-1} and the time base of 0.01 seconds, showing two full cycles of a 4 cm peak-to-peak sinusoidal

waveform that generates a loud alarm.

Task:

You have been approached as a student of Physics:

- a) Help the local community understand how the system works.
- b) Design a diagram showing how A.C from the solar inverter can be fully rectified to power the system.

TR. OPOLOT PATRICK.

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